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Past Papers

Nat 5

Physics

2024

Marking Scheme

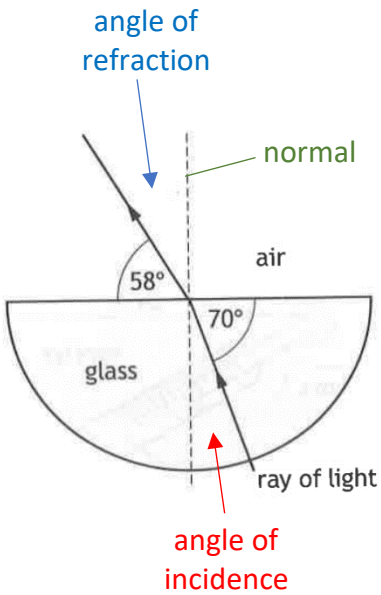
Grade Awarded	Mark Required		% candidates achieving grade
	/125	%	
A	86+	69%	34.4%
B	73+	58%	21.4%
C	60+	48%	18.0%
D	47+	38%	13.8%
No award	<47	<38%	12.3%

Section:	Multiple Choice	Extended Answer	Assignment
Average Mark:	16.0 /25	41.2 /75	17.386 /25

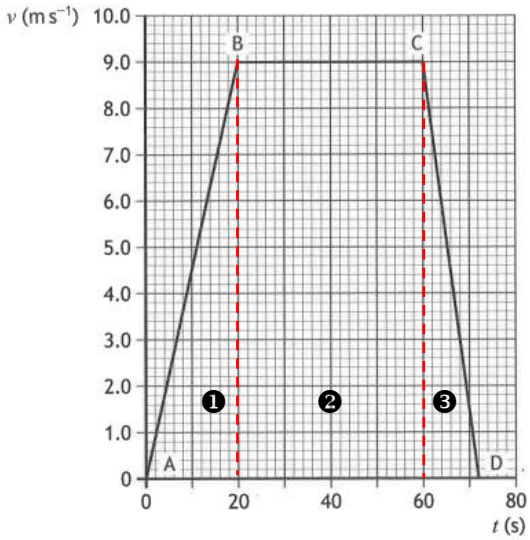
2024 Nat5 Physics Marking Scheme

Question	Answer	% Correct	Physics Covered
1	E	64	Vector Quantity: force, velocity, displacement, acceleration, **weight** Scalar Quantity: **energy**, **speed**, distance, time, mass
2	B	75	**Average speed between X and Y**: average speed = $\frac{\text{total distance}}{\text{time taken}}$; average speed = $\frac{d}{t_1}$ **Instantaneous speed at Y**: instantaneous speed = $\frac{\text{length of card}}{\text{time to pass light gate}}$; instantaneous speed = $\frac{l}{t_2}$
3	E	58	☒A The force of the Earth on exhaust gases is not the reaction force in example. ☒B The force of the Earth on the rocket engines is not the reaction force in example. ☒C The force of the rocket engines on Earth is not the reaction force in example. ☒D The force of the exhaust gases on Earth is not the reaction force in example. ☑E The force of the rocket engines on the exhaust gases is equal and opposite to the force of the exhaust gases on the rocket engines (Newton's Third Law)
4	B	64	E_p = ? m = 2400kg g = 9.8 N kg^{-1} h = 5.0m $E_p = m \quad g \quad h$ $E_p = 2400 \times 9.8 \times 5.0$ $E_p = 117600$ J E = 117600 J P = ? t = 2 minutes = 2x60s $P = \frac{E}{t} = \frac{117600}{2x60} = 980$ W
5	A	63	Statement I - Correct: The horizontal speed of the satellite is constant but not the velocity as it is not travelling in a straight line Statement II - Incorrect: The satellite is moving in both the horizontal and vertical directions to stay in orbit Statement III - Incorrect: The is still gravitational field strength while in orbit around Earth.
6	D	39	Time for laser light to travel to moon and back = 2.6s Time for laser light to travel to moon and back = 1.3s $d = v \times t$ $d = 3x10^8 \text{ m s}^{-1} \times 1.3s$ $d = 3.9x10^8$ m
7	C	91	AprizeSat-3 has a higher altitude (686km) than Hubble Space Telescope (537km) ∴ AprizeSat-3 must have period greater than 95 minutes AprizeSat-3 has a lower altitude (696km) than Intelsat-18 (35800km) ∴ AprizeSat-3 must have a period less than 1440 minutes
8	D	81	Statement I - Incorrect: A light year is the distance travelled by light in one year. Statement II - Correct: The approximate age of the universe is 13.8 billion years. Statement III - Correct: The Big Bang Theory describes the origin of the universe
9	A	47	☑A Particle is attracted to Q & R ∴ Particle has opposite charge to particles Q & R ☒B Particle is attracted to Q ∴ Particle and Q cannot have same charge ☒C Particle is attracted to R ∴ Particle and R cannot have same charge ☒D Particle is attracted to R ∴ Particle and R cannot have same charge ☒E Particle is attracted to Q ∴ Particle and Q cannot have same charge

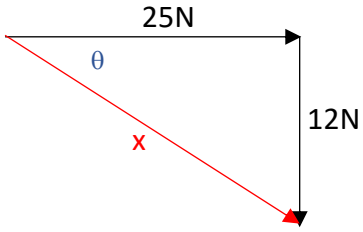
10	B	43	Statement I - Incorrect	Statement II - Correct	Statement III - Incorrect
			1 Watt of power is the transfer of 1 Joule of energy per second 1.2kW = 1200W = 1200 J s^{-1}	Appliances over 720W require a 13A fuse to be fitted to the plug.	P = 1.2kW = 1200W I = ? V = 230V P = I V 1200 = I x 230 I = $^{1200}/_{230}$ = 5.22A Q = ? I = 5.22A t = 1s Q = I t Q = 5.22 x 1 Q = 5.22C
11	E	63	V_1 = 3.6V V_2 = ? R_1 = 16Ω R_2 = 24Ω $\frac{V_1}{V_2} = \frac{R_1}{R_2} \therefore \frac{3.6}{V_2} = \frac{16\Omega}{24\Omega} \therefore \frac{3.6 \times 24}{16} = V_2 \therefore V_2 = 5.4V$		
12	A	28	Reading on Ammeter A_1	Reading on Ammeter A_2	Reading on Ammeter A_3
			Decreases	Decreases	Decreases
			As light on LDR decreases the resistance of LDR increases. Total resistance in circuit increases and current in A_1 will decrease as result.	As current in A_1 decreases, the split of this lower current at the branches will give a lower current in A_2	As current in A_1 decreases, the split of this lower current at the branches will give a lower current in A_3
13	B	65	$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} = \frac{1}{12} + \frac{1}{6.0} + \frac{1}{4.0} = \frac{1}{12} + \frac{2}{12} + \frac{3}{12} = \frac{6}{12} \therefore R_T = {}^{12}/_{6} = 2.0\Omega$		
14	C	86	P = ? I = 1.5A R = 8.0Ω $P = I^2 R$ $P = (1.5)^2 \times 8.0$ $P = (1.5)^2 \times 8.0$ $P = 2.25 \times 8.0$ P = 18 W		
15	B	84	☒A Heating block for longer increases temperature gained but not improve accuracy ☑B Insulation would reduce heat loss from block and improve accuracy ☒C A block of lesser mass would gain a higher temperature but not improve accuracy ☒D Higher power would increase temperature gained but not improve accuracy ☒E Removing heat proof mat leads to more energy being lost and decrease accuracy		
16	D	62	☒A P→Q represents the increase in temperature in the solid state ☒B Q→R represents the latent heat of fusion from solid to liquid ☒C R→S represents the increase in temperature in the liquid state ☑D S→T represents the latent heat of vaporisation from liquid to gas ☒E T→U represents the increase in temperature in the gas state		
17	D	52	$V_1 = 1.40 \times 10^{-6}\ m^3$	p_1 = 120 kPa	Temperature constant
			$V_2 = 1.40 \times 10^{-6}\ m^3 - 0.30 \times 10^{-6}\ m^3 = 1.1 \times 10^{-6}\ m^3$	p_2 = ?	
			$p_1 \times V_1 = p_2 \times V_2$ $1.40 \times 10^{-6} \times 120 = p_2 \times 1.1 \times 10^{-6}$ $\frac{1.40 \times 10^{-6} \times 120}{1.1 \times 10^{-6}} = p_2$ p_2 = 153 kPa		

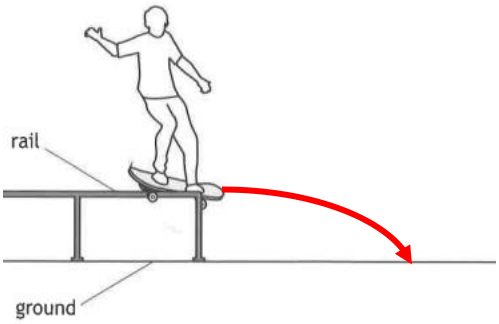
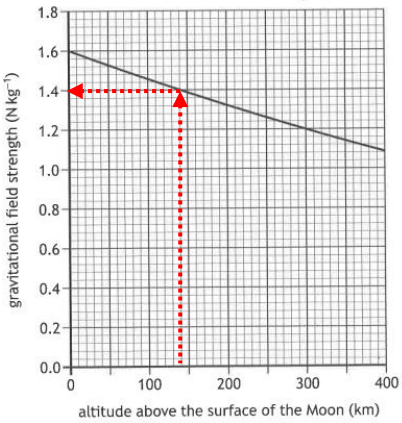
18	A	77	F_b = 360 N p = 1020 kg m^{-3} g = 9.8 N kg^{-1} V = ? $F_b = p \quad g \quad V$ $360 = 1020 \times 9.8 \times V$ $V = \frac{360}{1020 \times 9.8}$ $V = 0.036\ m^3$
19	C	74	3 waves travel 24m at 28 m s^{-1} therefore v = 48 m s^{-1} d = 24 m t = ? $v = \frac{d}{t} \quad \therefore \quad 48 = \frac{24}{t} \quad \therefore \quad t = \frac{24}{48} = 0.5\ s$ f = ? N = 3 t = 0.5 s $f = \frac{N}{t} = \frac{3}{0.5} = 6.0\ Hz$
20	C	57	<table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>**Longer**</td><td>**More**</td><td>**The Same As**</td></tr><tr><td>Infrared has a longer wavelength than visible light</td><td>Longer infrared wavelengths diffract more than shorter visible wavelengths</td><td>Infrared and visible light are both EM radiation and both travel at 3x10^8 m s^{-1}</td></tr></table>
21	A	64	Angle of Incidence = 90° – 70° = 20° Angle of Refraction = 90° – 58° = 32° 
22	B	79	☒A Removing a proton would be a nuclear reaction and not part of ionisation ☑B Ionisation involves the removal of an electron to form a charged particle ☒C Removing a (neutral) neutron would not change the charge on the particle ☒D Nuclear Fission: Splitting a large nucleus into smaller nuclei ☒ E Nuclear Fusion: Joining of small nuclei to form a larger nucleus
23	E	69	A = ? N = 1.8x10^6 nuclei decay t = 10 hours = 10 x 60 x 60 s $A = \frac{N}{t} = \frac{1.8x10^6}{10 \times 60 \times 60} = 50\ Bq$

24	C	63	$H = D\,w_r$ (alpha radiation) $+ D\,w_r$ (slow neutrons) $= (0.20\ \mu Gr \times 20) + (6.0\ \mu Gr \times 3)$ $= 4.0 +$ $= 18.6\ \mu Sv$
25	B	37	☒A Original reading of 400 counts s^{-1} has not halved to 200 counts s^{-1} by 20 minutes ☑B Increasing the time to 40 minutes would allow the count rate to fall below 200 counts per second allowing an easier measurement of half-life from the graph ☒C Lead shielding would stop/reduce the radiation reaching the Geiger-Müller tube ☒D Closer detector increases count rate but graph will not have halved in that time ☒E Further detector decreases count rate but graph will not have halved in that time

Question	Answer	Chemistry Covered
1a(i)	Constant Acceleration	When graph is uphill in a straight line — **constant acceleration**
1a(ii)	Constant speed	When graph is horizontal in straight line — **constant speed** When graph is downhill in a straight line — **deceleration** or **negative acceleration**
1b	- 0.75 m s^{-2}	a = ? v = 0.0 m s^{-1} u = 9.0 m s^{-1} t = 12 s $a = \frac{v - u}{t} = \frac{0.0 - 9.0}{12} = -0.75\ m\ s^{-2}$ (1 mark) (1 mark) (1 mark)
1c	504m	

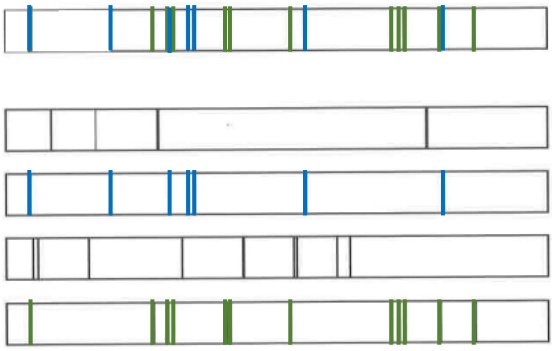
Area ❶	Area ❷	Area ❸
Distance = area under graph	Distance = area under graph	Distance = area under graph
= ½ x 20 x 9	= 40 x 9	= ½ x 12 x 9
= 90	= 360	= 54
Total distance = 90 + 360 + 54 = 504m		

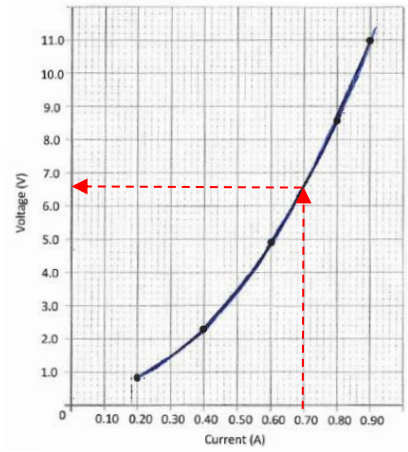
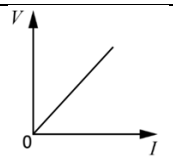
2a(i)A	28 N	 $x = \sqrt{(25)^2 + (12)^2}$ $x = \sqrt{625 + 144}$ $x = \sqrt{769}$ $x = 28$ N
2a(i)B	116	$\tan\theta = \frac{opp}{adj} = \frac{12}{25} = 0.48 \quad \therefore\ \theta = 26^o$ $\therefore$ bearing = 90° + 26° = 116
2a(ii)	0.37 m s^{-2} at 116	a = ? F = 28 N m = 75 kg F = m a (1 mark) 28 = 75 x a (1 mark) a = 0.37 m s^{-2} at 116 (1 mark)
2b	One answer from:	adopt a streamlined position / cycle behind another triathlete / wear tight clothing/lycra / solid wheels instead of spokes
2c(i)	4.4 m s^{-1}	d = 4 x 2.5 km = 4 x 2500 m = 10000 m $\bar{v}$ = ? t = 38 minutes = 38 x 60 s = 2280 s d = $\bar{v}$ t (1 mark) 10000 = $\bar{v}$ x 2280 (1 mark) $\bar{v}$ = $^{10000}/_{2280}$ = 4.4 m s^{-1} (1 mark)

2c(ii)	0 m s^{-1}	The total displacement after 4 laps back to the same point would be 0 m ∴ average velocity 0 m s^{-1}
3a(i)	1300J	At start: E_k = ? m = 65 kg v = 7.0 m s^{-1} $E_k = \frac{1}{2} m v^2$ $E_k = \frac{1}{2} \times 65 \times (7.0)^2$ E_k = 1592.5 J At end: E_k = ? m = 65 kg v = 3.0 m s^{-1} $E_k = \frac{1}{2} m v^2$ $E_k = \frac{1}{2} \times 65 \times (3.0)^2$ E_k = 292.5 J **Decrease in Kinetic Energy** = 1592.5 – 292.5 = 1300 J
3a(ii)	650 N	E_w = 1300 J F = ? d = 2.0m $E_w = F d$ (1 mark) $1300 = F \times 2.0$ (1 mark) F = 650 N (1 mark)
3b	Curved Arrow drawn as shown:	
4a	94 N	 From graph: When altitude = 140 km Gravitational field strength = 1.4 N kg^{-1} W = ? m = 67 kg g = 1.4 N kg^{-1} W = m x g (1 mark) W = 67 x 1.4 (1 mark) W = 94 N (1 mark)
4b	one answer from:	exposure to radiation; pressure differential; lower gravitational field strength; extreme temperatures
4c	answer to include:	1 mark: Acceleration increases 1 mark: mass decreases (as fuel is used) or gravitational field strength decreases (with altitude) 1 mark: weight decreases or unbalanced force increases
5	Open ended question:	1 mark: Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem. 2 marks: Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem. 3 marks: Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.
6a	one answer from:	not affected by weather; no (distortion from) atmosphere; no light pollution; can use telescope during the day; Full range of EM waves can be observed
6b(i)	A planet outside our solar system	Exoplanets orbit a star outside our solar system and information about the size and orbit od the star can be worked out from the amount of light arriving from the star as the planet crosses the sun

Candidate Marks gained:

3 marks	2 marks	1 mark	0 marks
5%	14%	37%	44%

6b(ii)	working showing 3.9×10^{17} m	d = ? $v = 3.0 \times 10^{8}$ m s^{-1} t = 41 years = 41 x 365.25x24x60x60 s d = v x t (1 mark) d = 3.0×10^{8} x 41 x 365.25 x 24 x 60 x 60 (1 mark) d = 3.9×10^{17} m (1 mark)
6c	Helium + Nitrogen	 Star Hydrogen [Not in star] Helium Mercury [Not in star] Nitrogen
7a	electrons flow in one direction (only)	d.c. current has electrons/negative charges flowing in one direction only. a.c. current has the direction of electron flow continually changing back and forth,
7b	13500 C	Q = ? I = 2.5A t = 1.5 hours = 1.5x60x60s Q = I t (1 mark) Q = 2.5 x 1.5x60x60 (1 mark) Q = 13500 C (1 mark)
7c	Answer to include:	1 mark (As the temperature increases) the voltage across the thermistor decreases 1 mark Voltage across the resistor increases 1 mark MOSFET turns on
7d	160Ω	Current in **series** circuit is same at all points = 18mA = 0.018A Total Voltage across both components in series = 5.0V Voltage across resistor = supply voltage – voltage across LED = 5.0V – 2.2 V = 2.8V V = 2.8V I = 0.018A R = ? V = I R (1 mark) 2.8 = 0.018 x R (1 mark) R = $^{2.8}/_{0.018}$ = 156Ω (1 mark)
8a(i)	Graph showing:	1 mark: suitable scales, labels and units 1 mark: all points plotted accurately to ± half a division 1 mark: best fit curve

8a(ii)	6.6 V	
8a(iii)	Vary the resistance of the variable resistor	The variable resistor can be used to increase and decrease the resistance in a circuit. As the resistance is decreased on the variable resistor, the current in the circuit would increase as a result due to Ohm's law. As the resistance of the variable resistor decreases, the voltage across the variable resistor would decrease meaning the voltage across the lamp would increase to maintain the same supply voltage at the battery.
8b		A fixed resistor has a constant resistance. This would give a straight line on the graph as the resistance (gradient) is constant. Line goes through the origin.
9a(i)	7800 J	E_h = ? c = 4180 J kg^{-1} $^oC^{-1}$ m = 0.020 kg ΔT= 100^0C – 6.3°C = 93.7°C E = c x m x ΔT (1 mark) E = 4180 x 0.020 x 93.7 (1 mark) E = 7833 J (1 mark)
9a(ii)	32000 J	E = ? m = 0.014 kg l = $22.6x10^5$ J kg^{-1} E = m x l (1 mark) E = 0.014 x $22.6x10^5$ (1 mark) E = 31640 J (1 mark)
9a(iii)	39800 J	Total Energy = 7800 J + 32000 J = 39800 J
9b(i)	140000 J	P = 750W E = ? t = 180s $P = \frac{E}{t}$ (1 mark) $750 = \frac{E}{180}$ (1 mark) E = 750 x 180 = 135000 J (1 mark)
9b(ii)	One answer from:	Heat loss to the surroundings. / Not all the energy used by the microwave is transferred to the water.
10a(i)	639 kPa	Volume: Constant p_1 = 655 kPa T_1 = 21°C = 294 K p_2 = ? T_2 = 14°C = 287 K (1 mark) $\frac{p_1}{T_1} = \frac{p_2}{T_2}$ (1 mark) $\frac{655}{294} = \frac{p_2}{287}$ $p_2 = \frac{655 \times 287}{294}$ (1 mark) p_2 = 639 kPa

10a(ii) — Answer to include:

1 mark	(As the temperature decreases)	the kinetic energy of the particles decreases particles move slower
1 mark	Particles collide with the walls of the container less frequently/often	or Particles collide with the walls of the container with less force
1 mark	Pressure decreases	

10b — 78.1 kg

$P = 1.02\times10^{6}$ Pa F = ? $A = 7.5\times10^{-4}$ m^2

$$P = \frac{F}{A} \quad \text{(1 mark)}$$

$$1.02\times10^{6} = \frac{F}{7.5\times10^{-4}}$$

$$F = 765 \text{ N}$$

W = 765N m = ? $g = 9.8$ N kg^{-1}

$$W = m \quad g \quad \text{(1 mark)}$$

$$765 = m \times 9.8 \quad \text{(1 mark)}$$

$$m = {}^{765}/_{9.8} = 78.1 \text{ kg} \quad \text{(1 mark)}$$

11a — Direction of vibration at right angles to the direction of wave travel/energy transfer

Wave Type	Description
Transverse	Direction of vibration at right angles to the direction of wave travel e.g. all electromagnetic waves and water waves
Longitudinal	Longitudinal waves have the direction of the vibration in the same direction as the direction of the wave e.g. sound waves

11b(i) — 0.14 Hz

t = 7.4s f = ?

$$t = \frac{1}{f} \quad \therefore \quad 7.4 = \frac{1}{f} = 0.14 \text{ Hz}$$

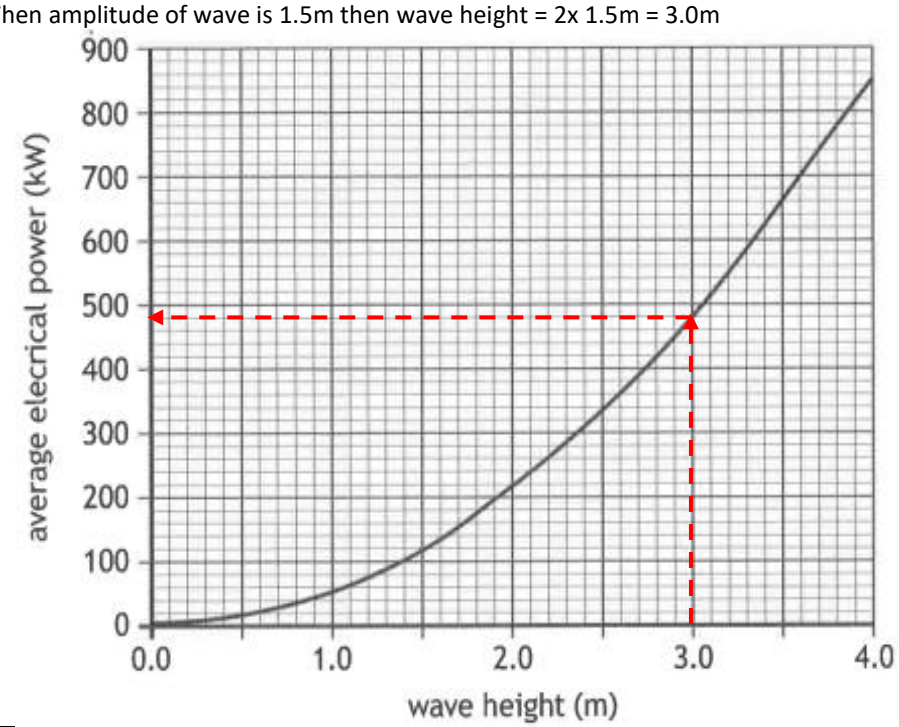
(1 mark) (1 mark)

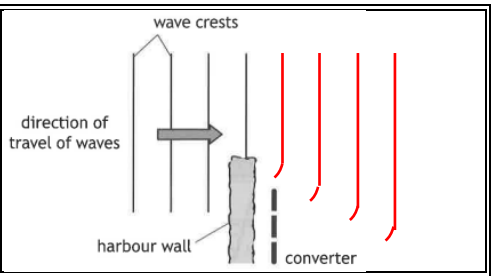
11b(ii) — one answer from:

measure (the time for) more waves to pass	count the number of waves in a longer period of time	repeat (the measurement) and average

11c — 480 kW

When amplitude of wave is 1.5m then wave height = 2x 1.5m = 3.0m



11d	Diagram showing:	<u>1 mark</u> diffraction of waves into right *shadow* region of the harbour wall <u>1 mark</u> consistent wavelengths before and after the harbour wall 
12a	The speed of sound (in air)	To calculate the distance, both the time and the speed/velocity are required. As lightning is so fast ($3x10^8$ m s^{-1}) that light is assumed to arrive simultaneous to the lightning flash.
12b(i)	One answer from:	ray of light is incident at { 90° / right angles / perpendicular } to the lens ray of light is travelling along the normal
12b(ii)	equal to	

Property	**Less Dense medium ↓ More Dense Medium** e.g. air → glass	**More Dense medium ↓ Less Dense Medium** e.g. glass → air
Wave Speed	deceases	increases
Change in Wavelength	deceases	increases
Change of Direction	bends towards the normal	bends away the normal
Frequency	**same**	**same**
Colour of Light	same	same

The frequency of the red light entering the lens is the same as it is inside the lens and also unchanged when it emerges out the other end of the lens.
The wave speed and wavelength would both decrease as the red ray enters the lens but would be back to the same values when it reemerges from the lens.

12c	Answer to include:	1 mark: (The amplitude of the second bang is) less than (the amplitude of the first bang) 1 mark: (The reflected sound has travelled further so) the second bang has less energy than the first bang

13 — Open ended question:

Candidate Marks gained:			
3 marks	2 marks	1 mark	0 marks
10%	25%	45%	21%

1 mark	2 marks	3 marks
Candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.	Candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.	Candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.